

SOURCE CODE:

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#include <stdio.h>

#include <stdlib.h>

int maxSpace = 0;

void mergeSort(int a[], int l, int r) {
    if (l >= r) return;

    int m = (l + r) / 2;
    mergeSort(a, l, m);
    mergeSort(a, m + 1, r);

    int n = r - l + 1;
    int *temp = malloc(n * sizeof(int));

    if (n > maxSpace)
        maxSpace = n;

    int i = l, j = m + 1, k = 0;

    while (i <= m && j <= r)
        temp[k++] = (a[i] <= a[j]) ? a[i++] : a[j++];

    while (i <= m) temp[k++] = a[i++];
```

```
while (j <= r) temp[k++] = a[j++];

for (i = 0; i < n; i++)

    a[l + i] = temp[i];

free(temp);
}
```

```
int main() {

    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int *a = malloc(n * sizeof(int));

    printf("Enter elements: ");
    for (int i = 0; i < n; i++)

        scanf("%d", &a[i]);

    mergeSort(a, 0, n - 1);

    printf("Sorted array: ");
    for (int i = 0; i < n; i++)

        printf("%d ", a[i]);
```

```
printf("\nMaximum auxiliary space: %d elements", maxSpace);  
printf("\nMemory used: %d bytes", maxSpace * (int)sizeof(int));  
  
free(a);  
return 0;  
}
```